

Assignment 1

CS402 – Parallel And Distributed Systems

Question 1:

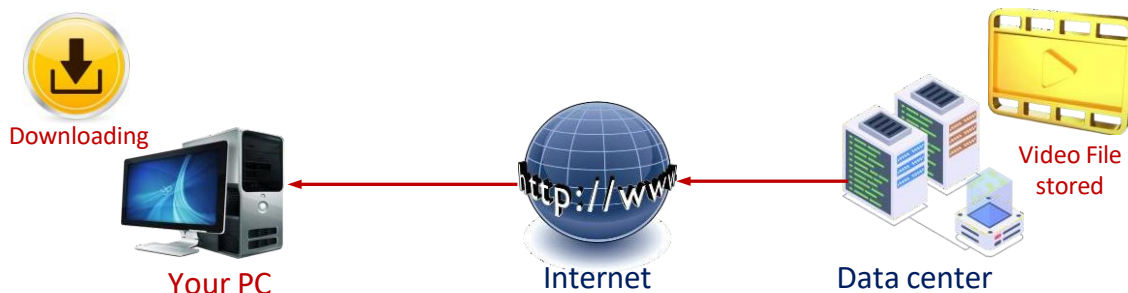
[10 Marks]

Describe and illustrate the microservices-based architecture of WeChat, Alipay, or similar large-scale Chinese applications, using block diagrams to explain your answer.

Question 2:

[10 Marks]

Imagine you're watching a video on your PC that traveled thousands of miles from a Google data center (as shown in the figure below). Describe and illustrate the video's (data) journey, mapping each step using the four layers of the TCP/IP protocol suite. Please clearly show each step.



Question 3:

[10 Marks]

A client sends a 200-byte request message to a service, which produces a response containing 5000 bytes. Estimate the total time required to complete the request in each of the following cases, with the performance assumptions listed below:

- using connectionless (datagram) communication (for example, UDP);
- using connection-oriented communication (for example, TCP);
- when the server process is on the same machine as the client.

[Latency per packet (local or remote, incurred on both send and receive): 5 ms

Connection setup time (TCP only): 5 ms

Data transfer rate: 10 Mbps

MTU: 1000 bytes

Server request processing time: 2 ms.

Assume that the network is lightly loaded.]

Question 4:**[10 Marks]**

Consider MIEC, Fuzhou University, is transitioning from IPv4 to IPv6 and needs to assign a subnet to a new residence hall with approximately 1,100 student devices. The university is allocated the prefix 2001:db8:1234::/48.

- Considering the need for additional subnets within the university in the future, determine a suitable subnet prefix length for the residence hall network that provides enough usable host addresses for the devices.
- Calculate the total number of usable host addresses available within this subnet.